



AUTOMATED QA WORKFLOW

Website Testing Agent

How the automated testing engine works — a plain-English guide for everyone. No technical background needed.

Deterministic Engine

No LLM · No API Keys

Headless Browser (Playwright)

Multi-Page Crawl

Evidence Screenshots

Excel / CSV Reports

This guide explains, in simple terms, how the Website Testing Agent works: how you enter a website address, how it opens the site in a real browser and walks through it like a manual QA tester, how it runs a fixed set of quality checks on every page and screen size, and how it reports each problem with a marked screenshot — all without any AI, API keys, or usage limits.

WHAT THIS GUIDE COVERS

- What the Website Testing Agent is, in plain words
- Why it is deterministic (and why that matters)
- The end-to-end flow: URL in → bug report out
- A step-by-step example of an audit running
- The full test-case toolbox (10 categories)
- How to read the report, evidence and exports

1 What is the Website Testing Agent?

The Website Testing Agent is an automated quality-assurance tool for websites. Instead of a person manually clicking through every page, checking spelling, resizing the browser, and writing bugs into a spreadsheet, you simply **enter a website address and click “Run audit.”** The agent opens the site in a real browser and inspects it the way a careful QA tester would.

It answers practical questions automatically, for example:

- “Are there any broken links or broken images on the site?”
- “Does the layout break on mobile, tablet, or small screens?”
- “Are there spelling mistakes, missing page titles, or missing meta descriptions?”
- “Do the contact forms validate input and keep the data when there’s an error?”

Every problem it finds becomes an **issue** with a severity, the exact page it’s on, steps to fix it, and a **screenshot with the problem area circled in red**. The results are shown live, organised page-by-page, and can be exported to Excel or CSV.

2 Why it is “deterministic” (not AI)

Some tools hand the work to an AI model that decides what to do on the fly. This agent does the opposite: it runs a **fixed, repeatable set of checks** written in code. The same website always produces the same results.

No LLM · No API keys · No rate limits

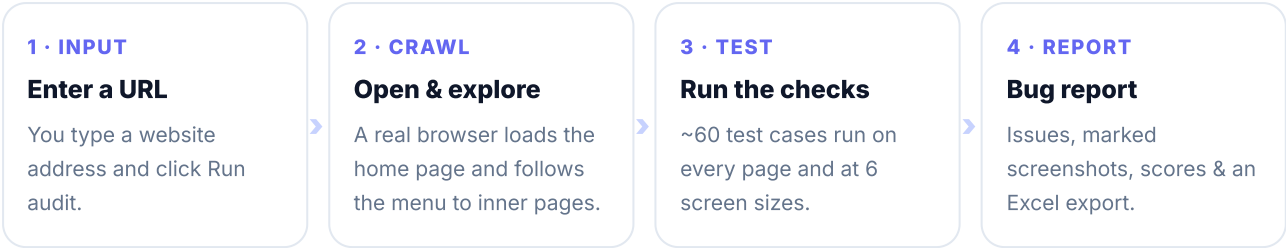
Because there is no AI model in the loop, audits never stall waiting on a quota, cost nothing per run, and are 100% repeatable. What you see is measured directly from the page — never guessed.

Deterministic vs. AI-driven — at a glance

	WEBSITE TESTING AGENT	AI-DRIVEN TOOL
Decides steps	Fixed checklist in code	Model decides each run
Repeatable	Yes — identical every time	Can vary between runs
Needs API key / quota	No	Yes
Evidence	Measured values + screenshots	Often summarised text
Cost per audit	None	Per-token / per-call

3 The end-to-end flow

Every audit moves through four stages. You give it one thing — a URL — and it returns a complete bug report.



What powers each stage

STAGE	WHAT HAPPENS
Browser	A headless Chromium browser (via Playwright) renders the site exactly like a visitor’s browser — including JavaScript, images, and fonts.
Crawl	The agent reads the navigation menu and footer, then visits up to several internal pages so the whole site is checked, not just the home page.
Checks	Small, single-purpose checks measure the page directly: link status codes, font sizes, heading order, form behaviour, image sizes, layout at each resolution, and more.
Evidence	For each problem the agent scrolls to the element, draws a red box around it, and captures a focused screenshot as proof.

Live progress

As the audit runs you see a live log stream and a progress ring in the app — every page visited and every problem found appears in real time over a streaming connection (Server-Sent Events).

WEBSITE TESTING AGENT — ARCHITECTURE & AUDIT FLOW

Deterministic Playwright QA engine · NO LLM · NO API KEYS · NO RATE LIMITS

1 INPUT

Website URL

https://yoursite.com

User enters a URL in the dashboard and clicks Run audit.

2 QA ENGINE (PLAYWRIGHT)

TestCaseAuditAgent

DETERMINISTIC · NO LLM

ENGINE RESPONSIBILITIES

- Launch headless Chromium
- Crawl home + menu/footer pages
- Run 60 test cases across 10 categories
- Capture per-page evidence screenshots
- Stream live logs & progress (SSE)

Aggregate Results

Combine every page's findings into issues, checklist status & scores — then return the report.

3 TEST-CASE ACTIONS (PER PAGE)

1

navigate & crawl
Visit each page, capture screenshot & DOM

PAGE LOAD

2

check_links
Broken links, redirects, anchors

HEAD / GET

3

scan_layout
6 resolutions: overflow, overlap, padding

VIEWPORT

4

check_content
Spelling, grammar, fonts, headings

DOM SCAN

5

test_forms
Labels, tooltips, validation, errors

INTERACT

6

check_seo
Meta title/description, H1, per page

DOM SCAN

4 EXTERNAL SOURCES

Target Website

GET — Home + crawled pages (HTML, DOM, CSS)
https://yoursite.com/#

Link Validation

HEAD/GET — Status of every internal & external link
redirect-aware · up to 120 Links

Page Assets

GET — Images, favicon, fonts (weight & alt-text)
resource-timing · transfer size

Social Profiles

GET — Validate footer social links resolve
Facebook · x · Instagram · LinkedIn

Browser Storage

LOCAL — Persist session (IndexedDB), survives refresh
no server · no cloud · private

RESULTS AGGREGATION

Issues

Per-page bugs with severity & steps

Checklist

60 checks: pass / fail / warning

Screenshots

Evidence captured per page

Scores

SEO · Performance · ATty · Best-practice

Page Map

Every crawled page & its findings

A. LIVE RESULTS (IN-APP)

Live Audit Stream

Progress bar

Issue Tracker

Progress bar

Page-by-Page Report

Progress bar

Score Dashboard

Progress bar

Live logs & progress streamed to the UI over Server-Sent Events (SSE)

B. EXPORT (DOWNLOAD)

X

✓ All issues, per page

✓ Severity & fix steps

✓ Checklist status

✓ Device / resolution

XLSX

CSV

LEGEND

— Data flow

--- Action (engine → page)

GET / HEAD HTTP request

— Network request

--- Results (page → engine)

LOCAL Browser-side only

HOW IT WORKS

- The engine is fully deterministic — **no LLM, no API keys, no rate limits**, so audits never stall.
- Each page is crawled, then 60 test cases run per page and at 6 screen resolutions (640×480 — 1920×1080, tablet, mobile).
- Every finding is attributed to its own page and carries that page's evidence screenshot.
- Results stream live to the UI and persist in the browser; the full report exports to Excel / CSV.

Fig. The full architecture & audit flow — input, engine, test-case actions, sources, and outputs.

Kleza Solutions — Website Testing Agent

testingagent · v3.0

4 How an audit runs (a real example)

You watch each step appear live in the audit stream as the engine works through the site.

Example: auditing a multi-page clinic website.

Step by step

- **Branding & header** — loads the home page, checks the logo is visible and links home, the menu exists, the favicon and page title are present.
- **Page crawl** — follows the menu to inner pages (About, Services, Contact...), checking each one has a title and a link back home.
- **Broken links & anchors** — collects every link on every page and checks each returns a valid response — flagging 404s, dead links, and bad anchors.
- **Content, spelling & fonts** — scans copy for spelling and repeated words, checks heading order, and flags inconsistent fonts.
- **Forms & validation** — fills the contact form with test data, submits, and checks labels, tooltips, required-field errors, email validation, and that typed data survives an error.
- **Buttons, keyboard & images** — checks buttons are a consistent size, not cut off or overlapping, keyboard-reachable, and that images load and have alt text.
- **Resolutions & responsive** — resizes to 6 screen sizes (640×480 → 1920×1080, tablet, mobile) and checks for overflow, overlap, missing padding, and horizontal scrollbars.
- **Performance & SEO** — measures load time per page, flags oversized images, and checks meta titles, descriptions, and H1 headings.
- **Mark evidence** — re-opens each affected page, draws a red box on the exact problem element, and captures a focused screenshot.

Honest by design

The agent only reports problems it can actually measure. Checks it genuinely can't verify automatically (for example, whether a confirmation email was delivered) are left as

pending

for a human — never guessed or marked as passed.

5 The test-case toolbox

The engine runs roughly **60 checks across 10 categories** on every audit. You never choose them — they all run automatically.

Branding & Header

Logo visible & links home, menu present, header contact info, favicon, page title on every page.

Navigation & Links

Menu items work, home link on every page, no broken links, external links open in a new tab, anchors resolve.

Content, Spelling & Fonts

Spelling, repeated words, no placeholder text, consistent fonts, heading sizes & hierarchy, alignment, lists.

Forms & Validation

Labels, tooltips, required-field errors, email/phone validation, error spelling, data preserved on error, dropdowns.

Buttons & UI Standards

Consistent size & style, text not cut off, not overlapped, keyboard-accessible.

Images & Media

Images load, no broken images, alt text present.

Social Media & Footer

Social links present & valid, open in a new tab, footer links, privacy/terms, contact info & copyright.

Responsive / Resolutions

Layout intact at 6 screen sizes, no horizontal scroll, viewport meta tag, mobile & tablet integrity.

Performance & Usability

Page loads within 3 seconds, images optimised, reasonable page weight.

SEO & Meta Tags

Optimised page title, meta description present, single H1 per page, search-friendly URLs.

Every page, every size

Content, SEO and layout checks run on **each crawled page**

— and layout is re-checked at every screen size — so a problem on the “Services” page at mobile width is reported separately from the home page at desktop width.

6 Evidence & screenshot marking

Every problem that lives on a specific element gets a **focused screenshot** — the engine scrolls to the element, draws a **red box** around it, and captures just that part of the page. No more hunting through a full-page image to find what's wrong.

- **Layout issues** are captured at the exact screen size where they break, with the overflowing or overlapping elements boxed.
- **Content, image, link, form, button and footer issues** are boxed on the element they affect.
- **Each issue gets its own screenshot** — no two issues share the same generic image.
- **Abstract issues** (e.g. "fonts differ across pages", "page weight too heavy") have no single spot to box, so they're reported without a screenshot rather than a misleading one.

7 Reading the results

Severity

Every issue is rated so you can fix what matters first:

Critical breaks core function, SEO, or trust **Major** significant degradation **Minor** polish / best-practice

Where the results appear

VIEW	WHAT IT SHOWS
Dashboard	The 10 categories with a live pass / fail / warning status and progress bars.
Live audit stream	Real-time log of every page visited and problem found, with a progress ring.
Page-by-Page Report	Every audited page as a section — its issues grouped beneath it, sorted by severity, with a thumbnail.
Issue Tracker	A spreadsheet-style table of every issue: page, description, severity, status, assignee — editable.
Excel / CSV export	The full report, downloadable for sharing or record-keeping.

Your results are saved

Audits are stored in the browser, so the report, issues and history survive a page refresh — no re-running needed.

8 Manual QA scenarios it covers

The engine was built from a standard manual-QA test sheet. These real-world scenarios are now checked automatically:

- Content is free of spelling & grammar errors
- Fonts are consistent as per the design
- Text is properly aligned
- Error messages are correct and match field labels
- Tooltip text is present for every field
- Fields are properly aligned
- Adequate spacing between labels, rows & errors
- Buttons use a standard format and size
- Clicking the logo loads the home page
- A home link exists on every page
- Social media links are valid
- No broken links or images
- The site works at multiple resolutions
- Smooth navigation between pages
- Pages load within 3 seconds
- Scrollbar appears only when required
- Filled data is preserved on a submit error
- A title displays on every page
- Fields & buttons are keyboard-accessible
- Dropdown text is not truncated by field size

In summary

- ✓ **Enter a URL** — no setup, dashboards, or training needed.
- ✓ **Trust the findings** — measured directly from the page, with a marked screenshot as proof.
- ✓ **Repeatable & free to run** — no AI, no API keys, no usage limits.
- ✓ **Whole-site coverage** — every crawled page, at every screen size, reported page-by-page.

In one line

You enter a website address, a real browser walks the whole site and runs ~60 quality checks at six screen sizes, and you get a clear, page-by-page bug report with marked-up evidence — exportable to Excel, with no AI in the loop.